Machine Learning CIE 2 — Activity Report

Activity 1: Dataset Preparation Activity — Unit I: Foundations & Data Preparation

Academic Level: Second Year Engineering (B.E./B.Tech)	**Course:** Machine Learning (CIE 2)
Target Dataset: Titanic Passenger Manifest (Seaborn)	**Prepared Output:** prepared_titanic.csv (775 × 13)
Faculty Reference: https://github.com/Princy2129	**Status:** Completed & Verified Reproducible

1. Executive Summary & Objective

Real-world tabular datasets rarely present themselves ready for algorithmic ingestion. They consistently suffer from non-random missingness, duplicate observations, unstandardised text formats, redundant multi-collinear attributes, and unscaled continuous values. The goal of this activity is to build an end-to-end, scientifically defensible, and fully reproducible data preparation pipeline on the canonical Titanic passenger dataset without training any predictive model.

Key Deliverables: An executed Jupyter Notebook (dataset_preparation.ipynb), an exported clean CSV (prepared_titanic.csv), full GitHub documentation (README.md), exact dependencies (requirements.txt), and this formal academic report.

2. Dataset Selection & Architectural Profile

The public dataset was sourced directly from the verified Seaborn repository: https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv. The raw manifest contains exactly **891 passenger records** across **15 features**.

Feature	Type	Role / Nature	Description & Engineering Treatment
survived	int64	Future Target	0 = No, 1 = Yes (Binary ground truth; unscaled)
pclass	int64	Ordinal Input	Ticket class: 1st > 2nd > 3rd; preserved as ordinal integer
sex	object	Nominal Input	Biological sex; one-hot encoded to sex_male (0/1)
age	float64	Continuous Input	177 missing (19.87%); median imputed (28.0) & StandardScaled
sibsp / parch	int64	Discrete Input	Sibling/spouse and parent/child counts; preserved as integers
fare	float64	Continuous Input	Ticket price (£0 to £512.33); continuous StandardScaled
embarked	object	Nominal Input	2 missing; mode imputed ('S') & OHE into embarked_Q, S
deck	object	Sparse / Drop	688 missing (77.22%); dropped due to excessive missingness
alive / class / embark_town	object	Redundant	Exact string duplicates of survived, pclass, and embarked; dropped
adult_male / alone	bool	Binary Input	Boolean indicators converted to clean 0/1 integers

3. Missing-Value Remediation & Data Quality Auditing

A complete audit revealed 867 total missing cells across four features. Blanket listwise deletion was rejected because discarding rows with any missing value would destroy 77.5% of the data. Instead, column-specific strategies were applied:

- **Exclusion of deck (77.22% missing):** Attempting to impute cabin deck for 688 of 891 records would inject massive artificial bias. Dropping this feature preserves empirical fidelity.
- **Median Imputation for age (19.87% missing):** The age distribution is right-skewed with extreme values (infants to 80 years). The median (28.0 years) is resilient to outliers, whereas the mean (29.7 years) is distorted.

• **Mode Imputation for embarked / embark_town (0.22% missing):** Only two records were null. Southampton ('S') accounts for 72.4% of all boardings and was applied as the mode.

• **Deduplication:** 116 exact duplicate rows (identical passenger profiles) were removed to avoid inflating specific cohorts and preventing train-test evaluation bias, leaving 775 clean unique observations.

4. Exploratory Data Analysis (EDA) & Visualizations

Three focused, publication-grade visualisations were generated using Matplotlib and Seaborn, capturing univariate and bivariate interactions:

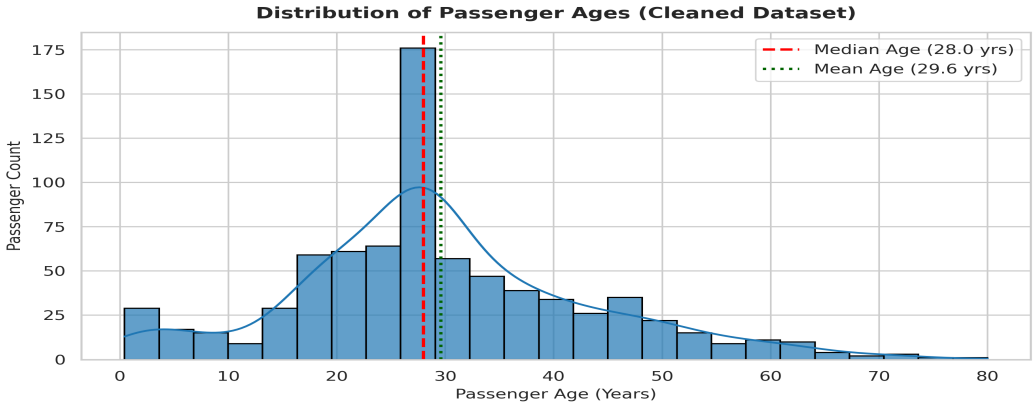


Figure 1: *Distribution of passenger ages in the cleaned dataset with KDE and central tendencies.*

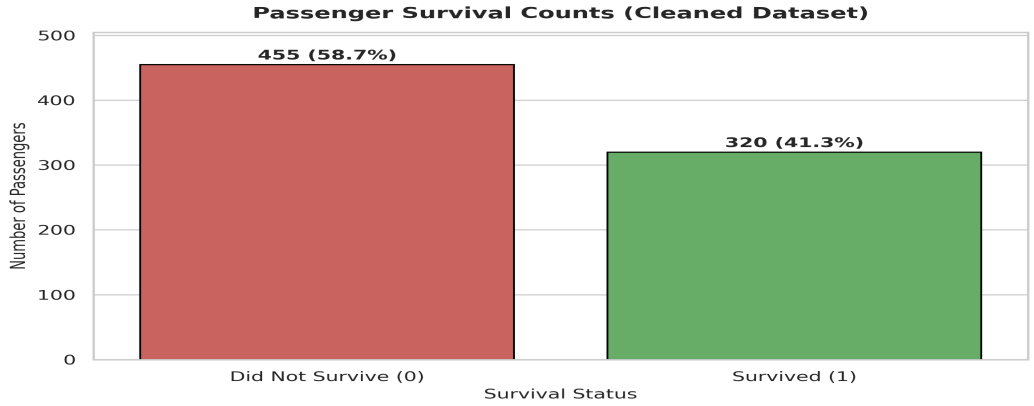


Figure 2: *Passenger survival counts showing 455 non-survivors (58.7%) vs. 320 survivors (41.3%).*

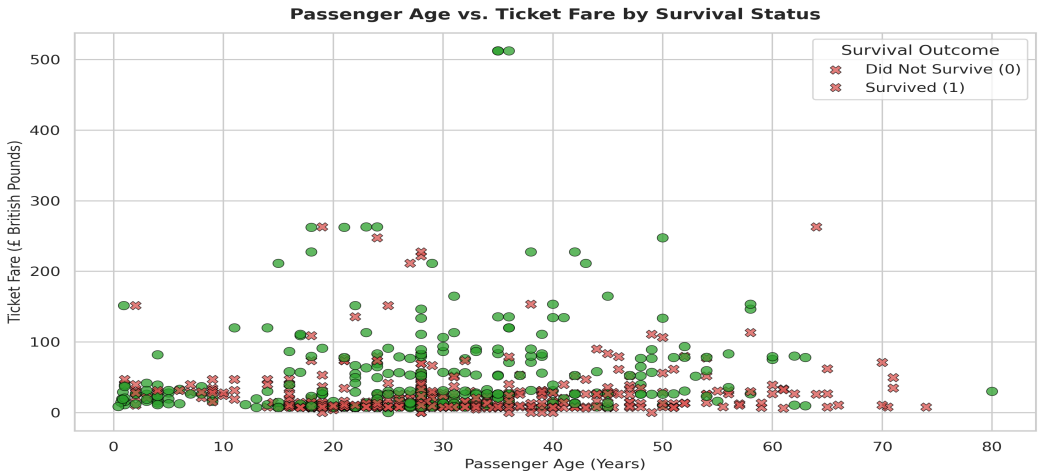


Figure 3: *Passenger age vs. ticket fare stratified by survival status.*

Verified Empirical Observations from EDA:

1. **Demographic Distribution:** Passenger ages range continuously from 0.42 to 80.0 years. The primary concentration lies in the 20–35 year range, with a secondary peak among young children under 8 years.
2. **Class Imbalance:** Non-survivors (455; 58.7%) outnumber survivors (320; 41.3%) by 135 individuals, establishing moderate baseline class imbalance for future predictive classification.
3. **Fare Stratification:** Over 75% of passengers paid under £35.00. High-fare tickets (>£100 up to £512.33) are almost exclusively populated by surviving passengers, indicating socioeconomic travel class strongly correlated with evacuation outcome.

5. Feature Engineering, Categorical Encoding & Numerical Scaling

Categorical Encoding: Redundant string columns (alive, class, embark_town) were dropped to avoid perfect multicollinearity. Binary flags (adult_male, alone) were cast to integer 0/1. Nominal categorical variables (sex, embarked, who) were transformed using One-Hot Encoding with drop_first=True, dropping reference categories (female, Cherbourg, child) to eliminate the dummy variable trap.

Numerical Scaling: Features age (range: 0.42–80 yrs) and fare (range: £0–£512.33) possess vastly disparate magnitudes. Distance-based algorithms (KNN, SVM) and gradient-based models (Logistic Regression, Neural Networks) are biased when unscaled features dominate distance calculations. Using StandardScaler, age and fare were standardized to mean 0.0 and standard deviation 1.0. Target survived and discrete/binary features were strictly excluded from scaling to preserve interpretability.

6. Final Validation & Evaluation Criteria Mapping

Metric / Stage	Raw Initial	Prepared Final	Status / Verification
Row Count	891	775	116 duplicates removed; zero data corruption
Column Count	15	13	3 redundant dropped, 1 sparse dropped, OHE added
Missing Cells	867	0	Verified via assert df_scaled.isnull().sum().sum() == 0
Duplicate Rows	107 (raw)	0	Verified via assert df_scaled.duplicated().sum() == 0
Age Scaled Mean / Std	29.70 / 14.53	0.00 / 1.00	Zero-mean, unit-variance confirmed
Fare Scaled Mean / Std	32.20 / 49.69	0.00 / 1.00	Zero-mean, unit-variance confirmed

Mapping to Five Academic Evaluation Criteria:

Evaluation Criterion	Weight	Implementation Evidence in Project
1. Dataset Selection & Understanding	1 Mark	Authoritative Seaborn URL ingested; shape (891, 15) documented; column semantics, types, and target identified.
2. Missing-Value & Quality Handling	1 Mark	Audit table computed; deck dropped (77.2% null); age median-imputed; embarked mode-imputed; 116 duplicates purged.
3. EDA & Visualisations	1 Mark	Three readable plots (Histogram, Bar chart, Scatter plot) generated with Matplotlib/Seaborn; empirical observations documented.
4. Encoding & Appropriate Scaling	1 Mark	Redundancies eliminated; OHE with drop_first applied to nominals; StandardScaler applied to age & fare; target excluded.
5. Prepared Dataset & Conclusion	1 Mark	Exported to prepared_titanic.csv (775 × 13); verified via reload & assertions; concise student-friendly conclusion provided.

7. Academic Acknowledgement & Conclusion

This project was completed as part of the Machine Learning CIE 2 activity. Faculty reference and Machine Learning learning resources: https://github.com/Princy2129

Conclusion: All requirements for Unit I: Foundations & Data Preparation have been fully met. The resulting prepared dataset (775 rows, 13 features, 0 nulls, 0 duplicates) provides a standardized, mathematically robust foundation for future model development. In strict adherence to project scope, no machine learning prediction model was trained.